



HORDE OF HORRORS

Comprehensive Project Breakdown & Strategic Roadmap

1. Executive Project Summary

Horde of Horrors is a portrait-oriented, mobile-first gothic horror wave survival game written in Godot 4.3 GDScript. The project is architected around highly optimized components targeting 60 FPS performance on standard mobile devices. Our development pipeline operates in two parallel tracks: **core client game engineering** and an **automated Grok-AI communication bridge** that allows us to query live conversation logs, receive optimized codebase proposals, and maintain detailed documentation.

2. Achievements & Completed Infrastructure

- **Autoload & Singleton Foundation:** Successfully registered core singletons in project.godot: UIManager (scene-based), PoolManager, GameManager, WaveManager, and JoystickLayer.
- **Virtual Joystick CanvasLayer (Joystick.tscn):** Designed and implemented a mobile portrait touch joystick that translates swipe inputs into global vectors.
- **Auto-Aim Target Scanning:** Added Area2D aim scanning to detect the nearest active monsters, enabling automatic weapon tracking essential for portrait gameplay.
- **Player Sprite & Logic State Machine:** Rebuilt PlayerController.gd and Player.tscn character body. Added clean movement, keyboard fallback, and automated state transitions (Walk, Run, Dash, Attack, Hurt, Die).
- **Core Scene Verification:** Rebuilt all 6 primary enemy scenes (Skeleton, Vampire, Werewolf, Ghost, Blood Golem, Plague Doctor) to conform to valid Godot 4.3 .tscn syntax.
- **Clean Damage Routing:** Linked EnemyBase.gd damage routines directly through the player's PlayerStatsComponent instead of bypassing standard handlers.
- **Interface Upgrades:** Refactored the game HUD to automatically hide during selection menus and transition smoothly into active gameplay. Resolved UpgradeShop naming conflicts in the canvas UI layers.
- **Automated Grok Integration:** Completed the Playwright/API synchronization bridge. All design logs are automatically retrieved, compared, and formatted in grok_conversation.md.

3. Current System Architecture: The Mini-Boss Framework

We are proceeding chronologically with **Week 2** (ranged combat/pooling) and **Week 3** (progression/upgrade cards) next. However, our latest synchronization with Grok has yielded the complete codebase blueprints for the **Week 4 Mini-Boss System** early. This allows us to review and pre-architect the force-field mechanics, high-stakes arenas, and rare tier rewards before we reach the Week 4 milestone.

Mini-Boss Spawning Loop (WaveManager.gd):

Spawns a specific mini-boss scene every 10th wave. Spawning a boss automatically scales down regular mob populations to maintain frame rates. The system registers the boss instance, tracks its health, and triggers a ForceFieldArena node that locks the player inside a radius around the boss until the entity is defeated.

The Four Mythic Mini-Bosses:

- **Alpha Werewolf (Wave 10):** Aggressive pathing, high-speed chase. Triggers a ForceField at 60% and 25% health, gaining 80% damage reduction for 6 seconds.
- **Vampire Matriarch (Wave 20):** Combines fast bat-dash maneuvers with summon calls. Triggers a shield at 50% health, summoning a bat swarm for protection.
- **Revenant Frankenstein (Wave 30):** High-health juggernaut. Triggers a shield at 65% health, amplifying chain lightning capabilities.
- **Lich High Priest (Wave 40):** Ranged summoner/caster. Activates a high-power damage shield at 70% health, spawning waves of skeleton minions.

4. Strategic Milestone Roadmap (Weeks 2 — 6)

Week 2: Ranged Combat & Pathfinding

- Silver Crossbow:** Implement weapon resource parameters (damage, velocity, fire rate templates).
- Projectile Pooling:** Expand PoolManager to handle silver bolts and impact particle effects to preserve mobile memory.
- Smart Pathfinding:** Integrate NavigationAgent2D on EnemyBase.gd to bypass walls and layout obstacles intelligently.

Week 3: Progression & Drops

- Sinister Upgrades:** Design UI upgrade cards (Offensive/Defensive/Utility panels) triggered post-wave clearance.
- World Pickups:** Implement item drops on monster death (Blood essence, temporary speed boosts, temporary invincibility bubbles).

Week 4: Character Roster & Environments

- Roster Expansion:** Add character presets for Elias Voss & Serena Nightshade.
- Advanced State Machines:** Implement werewolf dash and vampire bat transformations.
- Map Development:** Build portrait-optimized layouts for the Cathedral, Mansion, and Ruins.

Week 5: Polish & Game Juice

- Visual Feedback:** Create a camera screenshake shader for impacts.
- GPU Particles:** Implement GPU-accelerated blood splatters on enemy deaths.
- AI Optimizations:** Implement visibility-based processing to pause AI routines for entities off-screen.

Week 6: Save Systems & Release Preparation

- JSON Serialization:** Build local file storage for player upgrades, high-scores, and unlocked lore pages.
- Victor Van Helsing:** Implement the final premium unlockable character class.
- Export Builds:** Configure Android APK and iOS project builds for final QA testing.

Roadmap Summary Table:

Status	Milestone Categories	Tasks
Completed	Autoload System, Mobile Joysticks, Target Scan, Syntax Fixes, Damage Router, HUD toggles, Git/Grok Bridge	11 Tasks
Pending	Crossbow weapons, Bolt pooling, Smart navigation, Upgrade cards, Pickups, Roster heroes, Boss states, Shaders, Save system, Mobile Export	17 Tasks